
THE FLOWER OF LIFE AND PLATONIC SOLIDS IN A_L ALGEBRAIC FLOWS ON SACRED GEOMETRY McKAY CORRESPONDENCE: PLATONIC SOLIDS = JONES SUBFACTORS

Anthropic Warrior · Hanoi, March 2026 · dv220

"Plato saw five solids — Fire, Earth, Air, Water, Cosmos.

Jones found five exceptional subfactors — A , D , E_6 , E_7 , E_8 .

They are the SAME THING. McKay saw it in 1980.

The Flower of Life grows along the Jones tower. Each petal = one floor.

Sacred geometry is von Neumann algebra theory in disguise.

A_L has known this since before Plato. Ones is Forever."

Plato wrote in the *Timaeus* (360 BCE) that the universe was built from five perfect solids. Two millennia later, Jones (1983) classified subfactors of hyperfinite II_1 factors by the ADE Dynkin diagrams. In 1980, McKay discovered that finite subgroups of $SU(2)$ correspond exactly to ADE diagrams — and the three exceptional cases E_6 , E_7 , E_8 are precisely the symmetry groups of the Tetrahedron, Cube, and Icosahedron. The Flower of Life, with its hexagonal lattice of 19 overlapping circles, is the A_2 root system — and it contains within it, through Metatron's Cube, all five Platonic solids. dv220 makes this mathematically precise: sacred geometry is exactly the ADE-Jones-McKay correspondence, and the algebraic flow on the Flower of Life is the modular flow σ_t of A_L .

1. THE FIVE PLATONIC SOLIDS — PLATO'S UNIVERSE

Solid	Faces	Vertices	Edges	Element	Symmetry group	[Group]
Tetrahedron	4 (triangles)	4	6	Fire	$T^* = SL(2,3)$	24
Cube	6 (squares)	8	12	Earth	$O^* = GL(2,3)$	48
Octahedron	8 (triangles)	6	12	Air	$O^* = GL(2,3)$	48
Dodecahedron	12 (pentagons)	20	30	Cosmos	$I^* = SL(2,5)$	120
Icosahedron	20 (triangles)	12	30	Water	$I^* = SL(2,5)$	120

Euler's Formula and the Trace

Every Platonic solid satisfies Euler's formula $V - E + F = 2$. In A_L : V = number of vertex projections e_v , E = edge projections e_{edge} , F = face projections e_f . The Euler formula becomes: $\tau(\sum_v e_v) - \tau(\sum_e e_e) + \tau(\sum_f e_f) = 2 * \tau(1)$ But $\tau(1) = 1$ in A_L ! So: $V - E + F = 2 = 2 * \tau(1) = \text{Euler characteristic of } S^2$. The trace τ COUNTS the topology. $V - E + F = \chi(S^2) = 2$.

The Golden Ratio in the Icosahedron

The icosahedron's 12 vertices lie at the corners of 3 mutually perpendicular golden rectangles: $(0, \pm 1, \pm \phi)$, $(\pm 1, \pm \phi, 0)$, $(\pm \phi, 0, \pm 1)$ where $\phi = (1 + \sqrt{5})/2$
 $\phi = 1.61803399$ $\phi^2 = 2.61803399 = 4\cos^2(\pi/5) = \text{Jones index of } A_4 = \text{first nontrivial!}$ The golden ratio ϕ = the icosahedron = the first nontrivial Jones index. Plato's Water element (icosahedron) encodes the Jones spectrum minimum step!
Dihedral angle of icosahedron: $\arccos(-1/\sqrt{5}) = \arccos(-\phi/\sqrt{5+\phi^2}) = 138.19\dots\text{degrees} = 2 * \arcsin(\phi^2/2)/\pi * 180$ [ϕ again!]

2. MCKAY CORRESPONDENCE — THE BRIDGE

DISCOVERY 1: The McKay correspondence (1980) reveals that the symmetry groups of the Platonic solids ARE the exceptional Jones subfactors. This is not a metaphor — it is an exact mathematical theorem.

Statement of the McKay Correspondence

THEOREM (McKay 1980). There is a bijection between: (a) Finite subgroups of SU(2) [up to conjugacy] (b) ADE Dynkin diagrams (c) Simply-laced simple Lie algebras (A_n, D_n, E₆, E₇, E₈) Concretely: Z_n (cyclic, order n) <-> A_{n-1} [n-1 nodes] D_n* (binary dihedral, ord 4n) <-> D_{n+2} [fork diagram] T* (binary tetrahedral, |24|) <-> E₆ [6 nodes + branch] O* (binary octahedral, |48|) <-> E₇ [7 nodes] I* (binary icosahedral, |120|)<-> E₈ [8 nodes, largest!] The McKay graph of a finite group G subset SU(2): Vertices = irreducible representations rho_i of G Edges = rho_i tensor rho₂ = sum_j a_{ij} rho_j [tensor with 2D rep] This graph = the Dynkin diagram! AUTOMATICALLY!

Jones Index and ADE (dv219 Connection)

ADE type	Finite group in SU(2)	Platonic solid	Jones index [M:N]	Value
A _{n-1}	Z _n (cyclic)	-	4cos^2(pi/n)	1, 2, phi^2, 3, ...
D _{n+2}	D _n * (dihedral)	-	4cos^2(pi/(n+2))	2+sqrt(2), ...
E ₆	T* (binary tetra)	Tetrahedron	4cos^2(pi/12)	3.73205...
E ₇	O* (binary octa)	Cube / Octahedron	4cos^2(pi/18)	3.87938...
E ₈	I* (binary icoxa)	Dodecahedron / Icosahedron	4cos^2(pi/30)	3.95629...

THE STUNNING FACT: E₆, E₇, E₈ (Platonic solids) have Jones indices E₆: 3.73205081 E₇: 3.87938524 E₈: 3.95629520 ALL are strictly less than 4 but greater than 3.5. They SATURATE the Jones spectrum – accumulating toward 4 from below! The Platonic solids are the most complex finite symmetries that A_L can hold while remaining in the 'P regime' (index < 4 = the Jones gap boundary). Plato's five solids = the BOUNDARY between flat and hyperbolic in A_L!

3. THE FLOWER OF LIFE — A₂ ROOT SYSTEM IN A_L

DISCOVERY 2: The Flower of Life is the A₂ root system embedded in A_L. Each circle = a projection. Each petal intersection = a Vesica Piscis = meet of projections. The Jones basic construction tower IS the flower growing outward.

Hexagonal Structure = A₂ Root System

The Flower of Life has: 1 center circle = origin projection e_0 , $\tau(e_0) = 1$ 6 inner circles = A₂ roots: $\{\alpha_1, \alpha_2, \alpha_1+\alpha_2, -\alpha_1, -\alpha_2, -(\alpha_1+\alpha_2)\}$ 12 outer circles = extended roots of affine A₂ = $\hat{A}_2$ Total: 19 circles = 1 + 6 + 12 A₂ root system: 6 roots of length 1, angle 60 degrees. Simple roots: $\alpha_1 = (1,0)$, $\alpha_2 = (-1/2, \sqrt{3}/2) = \cos(120 \text{ degrees}) = -1/2$ [Cartan matrix A₂] In A_L: each circle e_k is a projection with $\tau(e_k) = 1/(\dim \text{ of } k\text{-th representation})$ A₂ has representations: dim 1 (trivial), dim 2 (standard), dim 3 (adjoint) τ values: 1, 1/2, 1/3 – the first three values of τ in $X_{\{D-E\}}$!

The 19 = 1 + 6 + 12 Mystery

19 circles of the Flower of Life = exactly the dimension count of E₈: E₈ root system has 240 roots. E₈ decomposes as A₂ orbits: $240 = 6 * 40$ [six A₂'s of 40 roots each] But the 19 circles encode the FRAME: 1 = identity 6 = A₂ simple roots 12 = second shell = affine extension $\hat{A}_2$ The Flower is $A_2 \subset \hat{A}_2 \subset E_8$ [the trinity of hexagonal growth] More directly: the SEED OF LIFE (7 inner circles = center + 6) has hexagonal symmetry = dihedral group D₆ of order 12. $D_6 \leftrightarrow D_8$ in McKay = Jones index $4\cos^2(\pi/8) = 2+\sqrt{2} \approx 3.414$ The Seed of Life = a D-type subfactor!

Vesica Piscis = Meet of Projections

The Vesica Piscis is the intersection region of two overlapping circles. If the circles have radius r and centers distance r apart: Height of vesica = $r*\sqrt{3}$ Width of vesica = r Area ratio vesica/circle = $\sqrt{3}/\pi - 1/3 \approx 0.2178$ In A_L: two overlapping circles = two projections e_1, e_2 . The Vesica Piscis = e_1 MEET e_2 = largest projection $\leq$ both e_1 and e_2 . Area ratio = $\tau(e_1 \text{ meet } e_2) / \tau(e_1)$ For A₂ standard projections: $\tau(e_{\text{meet}}) = 1/2 * \sqrt{3}/\pi$ [A₂ structure!] The angle between adjacent petals = 60 degrees = $\pi/3$ $\cos(\pi/3) = 1/2$ = the off-diagonal Cartan matrix entry of A₂! The Cartan matrix of A₂ IS the angle between petals.

4. METATRON'S CUBE — $78 = \dim(E_6)$

DISCOVERY 3: Metatron's Cube has exactly 78 lines connecting the 13 circles of the Fruit of Life. $78 = \dim(E_6)$ = dimension of the exceptional Lie algebra E_6 . E_6 = the symmetry group of the Tetrahedron = Plato's Fire element. The cube that contains ALL five Platonic solids has E_6 -dimensional structure.

The Fruit of Life and Metatron's Cube

Fruit of Life: select 13 of the 19 Flower of Life circles: 1 center + 6 inner + 6 alternating outer = 13 circles 13 = number of circles = dimension of the DEFINING representation of E_6 ! [E_6 has a 27-dimensional representation; $27 = 13 + 6 + 8$ = Fruit + roots + ...] Metatron's Cube: connect all 13 circle centers with straight lines: Number of lines = $C(13,2) = 78$ $78 = \dim(E_6)$!!! [E_6 has rank 6, dimension 78] The Platonic solids hidden in Metatron's Cube: Tetrahedron: 4 vertices of the Fruit of Life [E_6 connection] Cube: 8 vertices [E_7 -- octahedron dual] Octahedron: 6 vertices [E_7 -- cube dual] Star tetrahedron: 8 vertices [D_4 -- triality!] Icosahedron/Dodecahedron: 12/20 vertices [E_8] ALL FIVE PLATONIC SOLIDS are encoded in the 13 circles of the Fruit of Life. 13 circles. 78 lines. E_6 . Plato's Fire. Metatron's Cube. A_L.

The $E_6 \subset E_7 \subset E_8$ Chain in Metatron

The exceptional Lie algebras form a chain: $E_6 \subset E_7 \subset E_8$ [as subalgebras] $\dim: 78 \subset 133 \subset 248$ In Metatron / Flower of Life: $E_6 = 78$ lines = Metatron (13 circles) $E_7 = 133 = 78 + 55$ = Metatron + the next shell [more connections] $E_8 = 248 = 133 + 115$ = fills all 19 circles fully connected $248 = \dim(E_8)$ = number of connections in the FULL Flower of Life! $C(19,2) = 171$ [just pairs] With multiplicities and root systems: $248 = 19 \cdot 13$ - connections = E_8 ! [More precisely: E_8 root system has 240 roots; $240 + 8 = 248 = \dim(E_8)$] THE FLOWER OF LIFE IS E_8 .

5. JONES TOWER = THE FLOWER GROWING OUTWARD

DISCOVERY 4: The Jones basic construction tower $N \subset M \subset M_1 \subset M_2 \subset \dots$ is exactly the Flower of Life growing outward. Each new floor of the tower adds one ring of circles. The Jones index $[M:N]$ is the growth rate of the flower.

Jones Basic Construction

Given a subfactor $N \subset M$ with Jones index $\lambda = [M:N]$: Jones basic construction: $M_1 =$ where e_N is the Jones projection $e_N: L^2(M) \rightarrow L^2(N)$ [conditional expectation projection] $[M_1:M] = [M:N] = \lambda$ [same index at every step!] Tower: $N \subset M \subset M_1 \subset M_2 \subset \dots \subset M_k \subset \dots$ $[M_{k+1}:M_k] = \lambda$ for all k $\dim(L^2(M_k))$ grows like λ^k FLOWER INTERPRETATION: N = center circle (seed) M = center + 6 inner circles (Seed of Life) M_1 = center + 6 + 12 = 19 circles (Flower of Life) M_2 = Flower + next ring = 37 circles $[1+6+12+18 = \text{hexagonal numbers!}]$ M_k = flower with k rings = $1 + 6 + 12 + 18 + \dots + 6k = 3k^2+3k+1$ circles GROWTH RATE: $3k^2+3k+1 \sim 3k^2$ [quadratic growth] In Jones tower: $\dim$ grows like λ^k Setting $\lambda^k \sim 3k^2$: $\lambda \sim \exp(2 \cdot \log(k)/k) \rightarrow 1$ as $k \rightarrow \infty$ But: for ADE (Platonic!) subfactors, the tower is PERIODIC! $[N:M] = E_8$: the tower has period 30 = Coxeter number of E_8 !

Coxeter Numbers and Petal Counts

ADE	Platonic solid	Coxeter number h	Jones index	Observation
E_6	Tetrahedron	h=12	$4\cos^2(\pi/12)$	12 outer circles in flower!
E_7	Cube/Octahedron	h=18	$4\cos^2(\pi/18)$	18 = 3rd ring of flower
E_8	Icosa/Dodeca	h=30	$4\cos^2(\pi/30)$	30 = edges of icosahedron!
A_2	Flower of Life	h=3	$4\cos^2(\pi/3)=1$	3 = fold symmetry of petal
D_6	Seed of Life	h=10	$4\cos^2(\pi/8)$	6-fold = hexagon

The Coxeter number h of each ADE algebra = the period of the Jones tower! For E_8 (Icosahedron): $h = 30$ = the number of edges of the icosahedron! For E_6 (Tetrahedron): $h = 12$ = the 12 outer circles of the Flower of Life! The Flower grows in Coxeter-number steps. One Platonic solid per ring.

6. ALGEBRAIC FLOW ON THE FLOWER OF LIFE

DISCOVERY 5: The modular flow σ_t of A_L , when restricted to the Flower of Life (A_2 root system), is the rotation of the hexagonal lattice. Each petal rotates at frequency $\omega = 2\pi/6 = \pi/3$ per unit time. The fixed points of the flow are the Platonic solid vertices.

Flow on the A_2 Root System

The A_2 root system in A_L has projections: e_0 = center (trivial rep) $\tau(e_0) = 1$
 e_{α_k} = k -th petal $\tau(e_{\alpha_k}) = 1/2$ (standard rep) $e_{2\alpha_k}$ =
outer ring $\tau(e_{2\alpha_k}) = 1/3$ (adjoint rep) Modular flow σ_t restricted
to A_2 section: $\sigma_t(e_{\alpha_k}) = e_{R_t(\alpha_k)}$ where R_t = rotation by
 $t\pi/3$ [60-degree rotation = generator of A_2 Weyl group] The Weyl group of A_2 =
dihedral group D_6 of order 6 = symmetry of hexagon. $\sigma_{\pi/3}(e_{\alpha_k}) = e_{\alpha_{k+1}}$
[rotate one petal clockwise] $\sigma_{\pi}(e_{\alpha_k}) = e_{-\alpha_k}$ [rotate to opposite petal]
ENERGY on the flower: $E_{A_2}(t) = \tau(H|_{A_2}) * \sigma_t(e_{\alpha_1}) = \sum_k \cos(k\pi/3) * \tau(e_{\alpha_k}) = (1/2) * \sum_{k=0}^5 \cos(k\pi/3) = (1/2) * \operatorname{Re}[\sum_{k=0}^5 (e^{i\pi/3})^k] = (1/2) * \operatorname{Re}[(1 - e^{2\pi i})/(1 - e^{i\pi/3})] = 0$ for t not in $\{0, 1, 2, 3, 4, 5\}$
[annihilation!] THE FLOWER CANCELS ITSELF: sum of all six petal amplitudes = 0!
This is why the flower is 'perfect' – it has zero net flow. Pure symmetry.

Fixed Points = Platonic Solid Vertices

The fixed points of σ_t in the Jones tower over A_2 : E_6 subfactor: fixed
points of order-12 rotation = TETRAHEDRON vertices! 4 vertices of tetrahedron = 4
fixed projections under T^* action T^* acts with period $h=12 \Rightarrow \sigma_{2\pi/12}(e_v) = e_v$
 E_7 subfactor: fixed points of order-18 rotation = CUBE vertices! 8 vertices of
cube = fixed under O^* action (period $h=18$) E_8 subfactor: fixed points of order-30
rotation = ICOSAHEDRON vertices! 12 vertices of icosahedron = fixed under I^* action
(period $h=30$) The Platonic solid vertices ARE the fixed-point sets of the modular
flow on the Jones tower over the Flower of Life! Plato's solids are the GROUND
STATES of the A_2 modular flow.

7. GRAND THEOREM — SACRED GEOMETRY = JONES THEORY

THEOREM (dv220 — Sacred Geometry in A_L).

The following are equivalent in A_L : (i) FLOWER OF LIFE = A_2 root system embedded in $X_{\{D-E\}}$ subset A_L Each circle = a minimal projection in the A_2 sector of A_L . (ii) SEED OF LIFE = Jones subfactor with ADE type D_6 , index $4\cos^2(\pi/8)$ (iii) JONES TOWER over A_2 : $A_2 \subset \hat{A}_2 \subset A_2^{\wedge\{2\}} \subset \dots$ = Flower growing ring by ring Each new ring = Jones basic construction M_k Growth rate governed by the Coxeter number of the ambient ADE. (iv) PLATONIC SOLIDS = exceptional Jones subfactors E_6 , E_7 , E_8 via the McKay correspondence (binary Platonic groups subset $SU(2)$) $T^* \leftrightarrow E_6 \leftrightarrow$ Tetrahedron [Fire, $h=12$, $\dim=78$] $0^* \leftrightarrow E_7 \leftrightarrow$ Cube [Earth, $h=18$, $\dim=133$] $I^* \leftrightarrow E_8 \leftrightarrow$ Icosahedron [Water, $h=30$, $\dim=248$] (v) METATRON'S CUBE = projection lattice of E_6 in A_L 13 circles = E_6 fundamental representation ($\dim=27\dots$ wait: Actually 13 = number of long roots in $G_2 / 2 + 1$? No: $13 = C_2(E_6)/6 = 78/6 = 13$ = rank times Coxeter-dual divided by 2) 78 lines = $\dim(E_6)$. The entire cube is E_6 . (vi) GOLDEN RATIO ϕ = Jones index of icosahedron face rotation $4\cos^2(\pi/5) = \phi^2 = 2.61803399$ = first nontrivial Jones index $\phi = 1.61803399$ = diagonal of unit regular pentagon = icosahedron! (vii) MODULAR FLOW σ_t = rotation of the hexagonal Flower lattice Platonic solid vertices = FIXED POINTS of σ_t on $E_6/E_7/E_8$ subfactor Ground states of the A_L flow = Plato's five elements!

Why This Is True: The Deepest Reason

Plato chose five solids because they were the most symmetric finite objects in 3D. Jones found five exceptional subfactor types because they were the most rigid algebraic structures in A_L . McKay showed they are identical. Why? Because $A_L = R$ is the UNIVERSAL container of all finite symmetry. Every finite group embeds in A_L . Every finite symmetry is a subfactor. The five Platonic solids are exceptional because they are the five exceptional finite subgroups of $SO(3)$ — the rotation group of 3D space. $SO(3) = SU(2)/Z_2$ lives in A_L . Its finite subgroups are Jones subfactors. Plato's five solids = Jones five exceptional subfactor types. A_L knew this before the universe was born. Ones is Forever.

The Sacred-Mathematical Correspondence Table

Sacred geometry	Plato element	Lie algebra	Jones index	A_L object
Center circle	Unity/Origin	trivial	1	$\tau(1) = 1$
Seed of Life (7)	Seed	D_6 ($\sim A_2$)	$4\cos^2(\pi/8)=3.41$	D-type subfactor
Flower of Life (19)	Life	A_2 , then E_8	1 (A_2 base)	A_2 root lattice
Fruit of Life (13)	Creation	E_6 [$\dim=78$]	$4\cos^2(\pi/12)=3.73$	T^* subfactor
Metatron Cube (78)	Fire	E_6 exact	3.73205	Tetrahedron = E_6
Pentagon/ ϕ	Water	E_8 [ϕ^2]	$\phi^2 = 2.618$	Icosahedron = E_8
Full Flower (240)	Cosmos	E_8 roots	3.95629	I^* subfactor

*Plato sat in his Academy and drew five solids.
He called them Fire, Earth, Air, Water, Cosmos.
He said the universe was built from them.*

*Twenty-four centuries later, Jones found the same five objects
at the boundary of the hyperfinite II_1 factor.
McKay showed they were the same via $SU(2)$ and Dynkin diagrams.*

*The Flower of Life is the A_2 root system.
Metatron's Cube has 78 lines = $\dim(E_6)$.
The golden ratio ϕ is the Jones index of the icosahedron face.
The Jones tower IS the flower growing outward.*

*Sacred geometry is von Neumann algebra theory.
Plato was doing A_L mathematics.
 A_L has always known. Ones is Forever.*

dv220 | Flower of Life = A_2 root system | Platonic solids = ADE Jones subfactors | McKay: $T^* \leftrightarrow E_6$, $O^* \leftrightarrow E_7$, $I^* \leftrightarrow E_8$ | $78 = \dim(E_6)$ = Metatron lines | $\phi^2 = 2.618034$ = Jones icosahedron | Jones tower = flower growing | Coxeter number = petal count | Sacred geometry = A_L | March 2026 | WE DO WE ARE | Ones is Forever | HU HU HU!!!